

IN3190 - Digital Signal Processing: Mandatory Project

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Task 1: The Overview

a)

The provided dataset containing the HDF5 sensor time series was parsed. Figure 1 displays the geographic map of the event location and the respective recording stations.

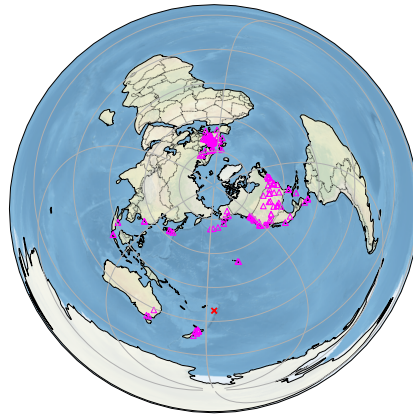


Figure 1: Geographic locations of the sensors and the Hunga Tonga volcano.

b)

Based on the station metadata, the calculated great circle distances from the Hunga Tonga event are as follows:

- **Shortest distance:** 2137.71 km
- **Longest distance:** 17687.13 km

c)

Figure 2 displays all sorted distances between the event location and the 201 station locations.

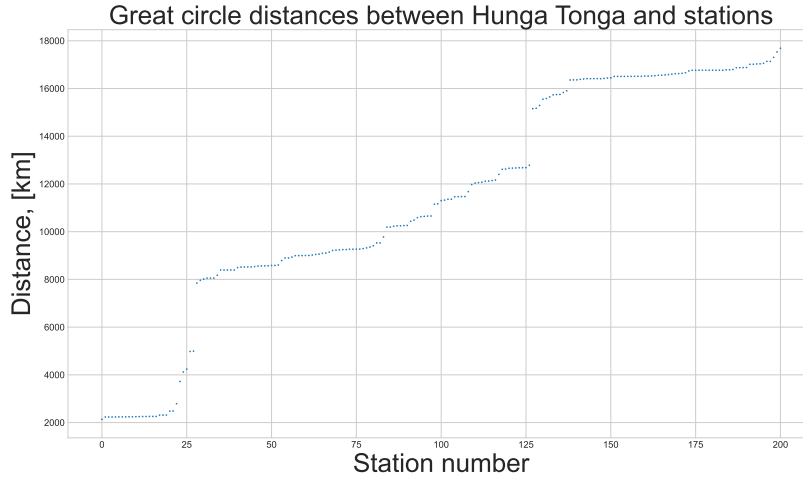


Figure 2: Sorted great circle distances from Hunga Tonga to all recording stations.

Task 2: Data Filtering

a)

The three provided FIR filter impulse responses, $h_1[n]$, $h_2[n]$, and $h_3[n]$, are plotted in Figure 3.

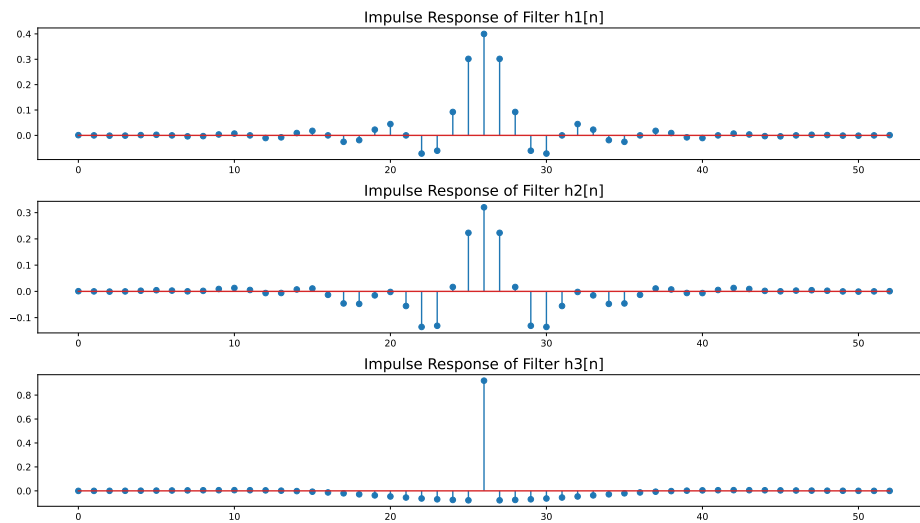


Figure 3: Impulse responses of the three FIR filters.

b)

See `task_2.py` for the implementation of the convolution function. The custom implementation is verified using `pytest` by directly comparing its output against NumPy's built-in `np.convolve` function.

c)

See `task_2.py` for the implementation of the DTFT function.

d)

The absolute values of the frequency spectra, $|H_1(e^{j\omega})|$, $|H_2(e^{j\omega})|$, and $|H_3(e^{j\omega})|$, are plotted against the physical frequency F in Figure 4.

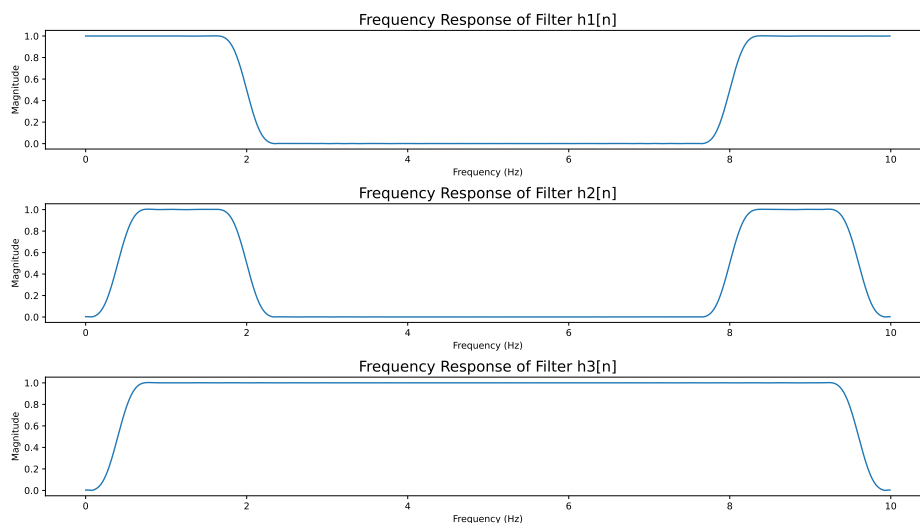


Figure 4: Absolute frequency spectra of the three provided FIR filters.

e)

Based on the frequency spectra observed in Figure 4, the filters are classified as follows:

- $h_1[n]$ is a lowpass filter.
- $h_2[n]$ is a bandpass filter.
- $h_3[n]$ is a highpass filter.

f)

All 201 input time series were successfully filtered using the three respective impulse responses via the built-in convolution function in `numpy`. See `task_2.py` for the implementation of the function that performs the filtering.

g)

from 4, it is evident that the bandpass filter $h_2[n]$ effectively isolates the frequency range of interest between 0.01 and 2 Hz. Therefore I chose to use this filter for further analysis.

Figure 5 shows the resulting section plot with time on the x-axis and distance from Hunga Tonga on the y-axis.

Figure 5: Section plot of the filtered waveforms (0.01–2 Hz) demonstrating global wave propagation.

Task 3: Celerity Estimation for the Acoustic Wave

a)

A custom routine was developed to loop over all station signals. For each station, the raw and filtered traces were analyzed to manually pick the arrival time of the higher-frequency infrasound

wave. The picked times were saved for subsequent analysis.

b)

Figure 6 plots the estimated travel times against the event-to-station distances. Identified outliers (caused by noise or competing arrivals) are marked in [Insert Color, e.g., red] and were excluded from the linear regression.

Figure 6: Linear fit of travel time versus distance for celerity estimation. Outliers are excluded from the fit.

By fitting a straight line to the accepted distance/time data, the slope yielded an estimated celerity of [Insert Value] m/s.

Comparing this result to the literature, this estimate aligns with typical celerities of infrasound waves as reported by Vergoz et al. (2022) and Matoza et al. (2022), which state [Insert brief comparison/typical values from the papers].